

E-Commerce Sales Analysis Report

Introduction

This project presents an analysis of an e-commerce dataset using Microsoft Power BI. The aim is to clean, model, analyze and visualize sales data to generate meaningful business insights that support decision-making.

The analysis was carried out using three main datasets:

- Customer dataset
- Sales dataset
- Product dataset

Power Query was used for data preparation, Power BI was used for modelling and visualization, DAX was used for creating analytical measures, and Python was used to create additional data visualizations.

Data Import and Cleaning

The datasets were imported into Power Query Editor where data quality checks and cleaning operations were performed.

The cleaning process included:

- Checking data types
- Ensuring numeric fields were correctly formatted
- Removing inconsistencies and errors where necessary
- Confirming relationships between tables

The cleaned datasets were loaded into Power BI Desktop for analysis.

Data Modelling

A relational data model was created using:

- Customer table
- Sales table
- Product table

The Sales table served as the main transaction table while Customer and Product tables acted as dimension tables.

Relationships created:

- Customer to Sales relationship
- Product to Sales relationship

This model enabled filtering and analysis across customers, products and sales transactions.

An additional Order table was imported from a MySQL database into Power BI to demonstrate database connectivity.

DAX Analysis

Fifteen DAX measures were created to generate business insights.

The measures included:

- Total Sales
- Total Orders
- Total Customers
- Total Quantity Sold
- Average Order Value
- Average Sales per Customer
- Maximum Sales
- Minimum Sales

- Product Count
- Average Product Price
- Quantity analysis measures
- Order Status
- Lowest Quantity Sold
- Highest Quantity Sold
- Category Sales

Dashboard Design and Insights

An interactive dashboard was developed containing cards, slicers and six visualizations.

- ✓ **Summary Cards:** Three cards were created:
 - Total Sales: Displays overall revenue generated.
 - Total Orders: Shows the number of completed transactions.
 - Total Customers: Shows the number of unique customers.
- ✓ **Slicer:** Three slicers were added to allow interactive filtering:
 - Product Category
 - Product Name
 - Customer/User ID

These slicers allow users to explore specific areas of the business data.

- ✓ **Visual Analysis**

I. Sales by Category (Donut Chart)

This visualization shows the contribution of each product category to total sales.

Insight:

Categories with larger portions represent stronger revenue contributors.

II. Top Products by Sales (Bar Chart)

This visual identifies the highest-performing products based on revenue.

Insight:

It helps identify products that contribute most to business income.

III. Quantity Sold by Product (Column Chart)

This shows products with the highest sales volume.

Insight:

High quantity products may represent popular items among customers.

IV. Average Product Price Analysis (Line Chart)

These visual compares product prices.

Insight:

It helps understand pricing differences among products.

V. Customer Sales Contribution (Pie Chart)

This visualization shows the contribution of top customers to total revenue.

Insight:

It identifies valuable customers who contribute significantly to sales.

VI. Product Performance (Treemap)

The treemap displays product contribution using size comparison.

Insight:

Larger blocks represent products generating higher sales.

Python-Based Data Visualizations

Python functionality in Power BI was used to create additional visualizations.

Four Python visuals were created:

I. Sales by Category (Bar Chart)

The chart compares total sales generated by each product category. Some categories contribute significantly more revenue than others, while some categories have relatively low sales performance.

Insight:

- High-performing categories represent the main revenue drivers.
- Low-performing categories may require promotional strategies or further investigation.

II. Top 10 Products by Sales (Horizontal Bar Chart)

This displays the top 10 products based on total sales value. The longer bars represent products generating higher revenue.

Insight:

- A small group of products contributes a large portion of total sales.
- These products can be prioritized for inventory management and marketing campaigns.

III. Sales Contribution by Category (Pie Chart)

The chart shows the percentage contribution of each category to overall sales.

Insight:

- Categories with larger slices have greater impact on revenue.
- It helps management identify the most valuable product segments and allocate resources effectively.

IV. Price vs Sales (Scatter Plot)

The scatter plot examines the relationship between product price and sales performance.

Insight:

- The points show that higher-priced products can generate higher sales, but price alone does not determine sales success.
- Some lower-priced products also perform well, suggesting that customer preference and product demand influence sales.

MySQL Integration

A table called order_items from a MySQL database was imported into Power BI using the MySQL connector.

This demonstrated Power BI's ability to connect with external relational database systems.

The imported order_items table was added into the Power BI model and relationship created with other existing tables.

Power BI Service Explanation

Power BI Service is the cloud-based platform used for publishing, sharing and managing Power BI reports.

It allows users to:

- Publish reports online
- Create dashboards
- Share insights with others
- Schedule data refresh
- Collaborate with teams

Power BI Desktop is mainly used for report creation, while Power BI Service is used for online sharing and distribution.

Conclusion

The E-Commerce Sales Analysis project demonstrates how Power BI can transform raw business data into meaningful insights. Through data cleaning, modelling, DAX calculations, interactive dashboards, Python visualizations and MySQL integration, the project provides a complete business intelligence solution for understanding sales performance, customer behaviour and product trends.